

The Full Saddle–Transseries Expansion of the Bell Numbers

Lambert– W sectors and combinatorial formulae for every coefficient

Prepared for the ProveIt research archive

September 2026

Abstract

Let B_n be the n th Bell number and let $r = W_0(n)$, so that $re^r = n$. We derive a complete fixed-order asymptotic expansion about the principal Cauchy saddle,

$$B_n = \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}} \left\{ \sum_{m=0}^{M-1} C_m(r) \left(\frac{r}{n}\right)^m + \mathcal{O}_M\left(\left(\frac{r}{n}\right)^M\right) \right\},$$

and give four equivalent, finite descriptions of every coefficient $C_m(r)$: a Gaussian coefficient extractor, an exponential-Bell-polynomial formula, a triangular recurrence, and an explicit sum over the integer partitions of $2m$. In particular, $C_m(r) = P_m(r)/\{r^m(1+r)^{3m}\}$ with $P_m \in \mathbb{Q}[r]$ and $\deg P_m \leq 4m$. The first five coefficients are written out, and the remainder is proved by a central-arc expansion together with a uniform complementary-arc estimate.

We then interface this expansion with Stirling’s series, derive an all-order factorial-free form and an all-order logarithmic (cumulant) form, and identify $\lim_{r \rightarrow \infty} C_m(r)$ with the reciprocal Stirling coefficients. The Lambert function is itself expanded on every branch by explicit Stirling–first-kind polynomials, converting the answer into the canonical nested $\log n$ – $\log \log n$ transseries.

Finally, using the Hankel representation with $N = n + 1$, we retain every contributory steepest-descent saddle $w_k = W_k(N)$. Each thimble sector is expanded to arbitrary order, with a second finite partition formula for its coefficients. The original Bell contour fixes the Stokes multipliers, while the singulants show that the $k \neq 0$ sectors are exponentially small like $\exp\{-2\pi^2 k^2 N / W_0(N)^2\}$ for fixed k . Thus the result includes both the complete perturbative series and the beyond-all-orders Lambert– W sectors, with explicit algorithms and numerical checks.

Scope and truth status

The word *full* is used in the following precise sense. First, every coefficient in the principal Poincaré expansion is given by a finite formula. Second, every complex Lambert– W saddle has an all-order local expansion with a finite coefficient formula. Third, the Bell contour selects a finite set of active thimbles at each fixed value of the large parameter, and this selection is stated explicitly in terms of Stokes data. Fourth, every Lambert branch can be re-expanded into the polynomial–logarithmic transseries scale.

The principal fixed-order error estimate in this article is analytic and uniform for sufficiently large n (with the truncation order fixed). The multisaddle expansion is sectorwise: no claim of a single error bound uniform in a branch index growing with n is made. At a Stokes threshold, lateral or median interpretation is required. The derivations are human-readable and computer-algebra checked, but are not claimed to be machine-verified in a proof assistant.

Contents

1 Introduction

3

1.1	Summary of the principal result	3
1.2	What is new in the organization	4
2	Exact representations and notation	4
2.1	Bell and Touchard polynomials	4
2.2	The circle representation	5
2.3	The Hankel representation and the shift convention	5
2.4	Asymptotic notation	5
3	The principal circle saddle to all orders	5
3.1	Exact local phase	5
3.2	Main fixed-order theorem	6
4	Combinatorial coefficient calculus	7
4.1	The Gaussian functional	7
4.2	Rational structure and degree bounds	8
5	Explicit principal coefficients	9
6	The Stirling interface	9
6.1	The limiting coefficients	9
6.2	Factorial-free all-order expansion	10
7	The logarithmic transseries	11
8	Lambert-W polynomial-logarithmic re-expansion	12
8.1	Every branch	12
8.2	Canonical nested-log form of $\log B_n$	13
9	The full Lambert-branch saddle transseries	14
9.1	Local data at an arbitrary saddle	14
9.2	All coefficients in one thimble sector	14
9.3	Thimble selection and Stokes multipliers	15
9.4	Relation between the two principal normalizations	16
10	Singulants and beyond-all-orders ordering	16
10.1	Exact singulant	16
10.2	Fixed-branch expansion	16
11	Executable coefficient generation	17
11.1	Stable numerical evaluation	19
12	Numerical validation	19
13	Proof-audit and formalization map	19
13.1	Analytic kernel	19
13.2	Algebraic kernel	20
13.3	Transseries kernel	20
14	Conclusions	20
A	A sharper view of the contour remainder	21

B Coefficient identities in expanded form	21
C Notation concordance	22

1 Introduction

The Bell number B_n counts the set partitions of an n -element labelled set. Its exponential generating function is

$$\exp(e^z - 1) = \sum_{n \geq 0} B_n \frac{z^n}{n!}. \quad (1)$$

The coefficient problem is classical, but its complete asymptotic structure is richer than the familiar leading formula suggests. The positive saddle moves to infinity with n and is determined implicitly by

$$re^r = n, \quad r = W_0(n).$$

Consequently the natural small parameter is not $1/n$ alone, but

$$q := e^{-r} = \frac{r}{n} \sim \frac{\log n}{n}.$$

At the same time a Hankel representation exposes infinitely many complex saddles $W_k(n+1)$; a finite, contour-dependent subset contributes for each fixed n . The nonprincipal contributions lie beyond every power of q and therefore belong to a genuine transseries rather than to the principal Poincaré series.

The classical analyses of Moser–Wyman, Hayman, de Bruijn, and later authors establish the leading shape and various refinements. Paris’s steepest-descent analysis of the Touchard polynomials makes the Lambert-branch saddle geometry and its Stokes changes explicit. Hwang identified a striking limiting connection between saddle coefficients and Stirling’s formula. The purpose of the present article is to assemble these ingredients into a self-contained coefficient calculus that satisfies five obligations simultaneously:

1. identify the actual asymptotic scale;
2. give a finite, executable formula for every coefficient;
3. state the exact shift convention (n versus $n+1$);
4. prove the central-arc remainder estimate and a complementary-arc bound;
5. include the exponentially small Lambert-branch sectors and their Stokes selection.

These are also the obligations singled out in the audit notes of the *Combinatorial Coefficient Calculus* draft in the ProveIt repository. The formal/analytic separation and the branch-aware logarithmic expansions follow the conventions of the companion *series-and-transseries* drafts.

1.1 Summary of the principal result

For orientation, define the leading saddle approximation

$$L_n(r) := \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}}, \quad r = W_0(n). \quad (2)$$

Then for every fixed $M \geq 1$,

$$B_n = L_n(r) \left(\sum_{m=0}^{M-1} C_m(r) q^m + \mathcal{O}_M(q^M) \right), \quad q = \frac{r}{n}. \quad (3)$$

The coefficient C_m is a rational function with a tightly controlled denominator. If $\lambda = (1^{m_1} 2^{m_2} \dots)$ is a partition of $2m$, write $\ell(\lambda) = \sum_j m_j$. Let

$$T_j(x) := \sum_{a=0}^j \left\{ \begin{matrix} j \\ a \end{matrix} \right\} x^a \quad (4)$$

be the j th Touchard polynomial. The finite coefficient formula is

$$C_m(r) = \sum_{\lambda \vdash 2m} \frac{(-1)^{m+\ell(\lambda)} (2m + 2\ell(\lambda) - 1)!!}{[r(1+r)]^{m+\ell(\lambda)}} \prod_{j \geq 1} \frac{T_{j+2}(r)^{m_j}}{m_j! ((j+2)!)^{m_j}}. \quad (5)$$

For $m = 0$ the empty partition gives $C_0 = 1$. Formula (5) is the central combinatorial object of the article.

1.2 What is new in the organization

No claim is made that every ingredient below is historically new. The contribution is the explicit synthesis and normalization:

- the circle-saddle expansion is placed directly in the scale $q = r/n$, with a finite partition formula and a fixed-order remainder;
- the denominator and degree structure of all C_m is proved;
- the Stirling, logarithmic, and polynomial–logarithmic interfaces are made algorithmic;
- the same partition calculus is transported to all Hankel thimbles, yielding an explicit full saddle transseries;
- the relation between the $r = W_0(n)$ and $w = W_0(n+1)$ normalizations is kept visible, avoiding a common off-by-one ambiguity.

2 Exact representations and notation

2.1 Bell and Touchard polynomials

The Touchard polynomials are

$$T_n(x) = e^{-x} \sum_{j \geq 0} \frac{j^n x^j}{j!} = \sum_{a=0}^n \left\{ \begin{matrix} n \\ a \end{matrix} \right\} x^a, \quad (6)$$

so $B_n = T_n(1)$. Their exponential generating function is

$$\exp(x(e^z - 1)) = \sum_{n \geq 0} T_n(x) \frac{z^n}{n!}. \quad (7)$$

The same polynomials occur as logarithmic derivatives of the exponential:

$$(xD_x)^j e^x = e^x T_j(x), \quad D_x := \frac{d}{dx}. \quad (8)$$

This identity is exactly what converts the local saddle derivatives into Stirling-number data.

The first few polynomials are

$$\begin{aligned} T_0(x) &= 1, & T_1(x) &= x, \\ T_2(x) &= x^2 + x, & T_3(x) &= x^3 + 3x^2 + x, \\ T_4(x) &= x^4 + 6x^3 + 7x^2 + x, & T_5(x) &= x^5 + 10x^4 + 25x^3 + 15x^2 + x. \end{aligned}$$

2.2 The circle representation

Cauchy's coefficient formula applied to (1), on the circle $z = re^{i\theta}$, gives the exact identity

$$\frac{B_n}{n!} = \frac{\exp(e^r - 1)}{2\pi r^n} \int_{-\pi}^{\pi} \exp(\Phi_r(\theta)) d\theta, \quad (9)$$

where

$$\Phi_r(\theta) := e^{re^{i\theta}} - e^r - in\theta. \quad (10)$$

The saddle equation $\Phi'_r(0) = 0$ is $re^r = n$. Throughout the principal analysis we therefore set

$$r := W_0(n), \quad q := e^{-r} = \frac{r}{n}, \quad A := n(1+r) = \frac{r(1+r)}{q}. \quad (11)$$

Here $A = -\Phi''_r(0)$ is the Gaussian curvature.

2.3 The Hankel representation and the shift convention

A second representation is needed to see all complex saddles. Put

$$N := n + 1. \quad (12)$$

By inserting a Hankel formula for the powers in Dobinski's sum, or directly from (7), one obtains

$$B_n = T_{N-1}(1) = \frac{\Gamma(N)e^{-1}}{2\pi i} \int_{-\infty}^{(0+)} \exp(N\psi_N(t)) dt, \quad \psi_N(t) := \frac{e^t}{N} - \text{Log } t. \quad (13)$$

The loop starts at $-\infty$, encircles the origin counterclockwise, and returns to $-\infty$; the logarithm is cut along the negative real axis. Its saddles satisfy

$$te^t = N, \quad w_k := W_k(N), \quad k \in \mathbb{Z}. \quad (14)$$

The shift in (12) is essential: the circle normalization uses $W_0(n)$, while the Touchard/Hankel normalization for $B_n = T_n(1)$ uses $W_k(n+1)$.

2.4 Asymptotic notation

All $\mathcal{O}_M$ constants may depend on the fixed truncation order M but not on n . Unless a branch index is explicitly allowed to vary, a statement about w_k is for fixed k as $N \rightarrow \infty$. Square roots in complex saddle formulae are fixed by continuous transport along the oriented steepest-descent thimble, with the conjugacy convention $\mathcal{J}_{-k} = \overline{\mathcal{J}_k}$ for $N > 0$.

3 The principal circle saddle to all orders

3.1 Exact local phase

Differentiating (10) and using (8) gives

$$\Phi_r(\theta) = -\frac{A}{2}\theta^2 + \sum_{j \geq 3} \frac{e^r T_j(r)}{j!} (i\theta)^j. \quad (15)$$

Scale

$$\theta = \frac{u}{\sqrt{A}} = \sqrt{\frac{q}{r(1+r)}} u. \quad (16)$$

For $j \geq 3$ define

$$\alpha_j(r) := \frac{T_j(r)}{j! [r(1+r)]^{j/2}}. \quad (17)$$

Then

$$\Phi_r\left(\frac{u}{\sqrt{A}}\right) = -\frac{u^2}{2} + \sum_{j \geq 3} \alpha_j(r) (iu)^j q^{(j-2)/2}. \quad (18)$$

Since $T_j(r)$ has nonnegative coefficients and degree j ,

$$0 \leq \alpha_j(r) \leq \frac{B_j}{j!} \quad (r \geq 1), \quad (19)$$

so every fixed family of normalized derivatives is uniformly bounded.

3.2 Main fixed-order theorem

Theorem 3.1 (Complete principal expansion). *Let $M \geq 1$ be fixed, $r = W_0(n)$, and $q = r/n$. As $n \rightarrow \infty$,*

$$B_n = \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}} \left\{ \sum_{m=0}^{M-1} C_m(r) q^m + R_M(n) \right\}, \quad (20)$$

where

$$R_M(n) = \mathcal{O}_M(q^M). \quad (21)$$

More precisely, the part of the contour outside a fixed small neighbourhood of $\theta = 0$ is

$$\mathcal{O}\left(\exp\left\{-c\frac{n}{r^2}\right\}\right) \quad (22)$$

relative to the unscaled central integral, for some absolute $c > 0$. The coefficients $C_m(r)$ are given in Theorem 4.1 below.

Proof. We give the proof in enough detail to identify the uniformity. Write $t = q^{1/2}$ and choose a fixed sufficiently small $\eta > 0$. Split the circle into three regions:

$$\begin{aligned} \mathcal{C}_0 : |u| &\leq t^{-1/4}, \\ \mathcal{C}_1 : t^{-1/4} &< |u| \leq \eta\sqrt{A}/r, \\ \mathcal{C}_2 : \eta/r &< |\theta| \leq \pi, \end{aligned}$$

where $u = \sqrt{A}\theta$. Because $\sqrt{A}/r \rightarrow \infty$, these regions are well ordered for large n .

On $\mathcal{C}_0$, (18) and (19) imply that the nonquadratic exponent is $\mathcal{O}(t|u|^3) = \mathcal{O}(t^{1/4})$. Expanding its exponential through degree $2M - 1$ in t gives

$$\exp\left(\sum_{\ell \geq 1} h_\ell(u; r) t^\ell\right) = \sum_{\nu=0}^{2M-1} P_\nu(u; r) t^\nu + \mathcal{O}_M\left(t^{2M}(1+|u|)^{6M} e^{u^2/4}\right), \quad (23)$$

with h_ℓ and P_ν defined in (28)–(29). After multiplication by $e^{-u^2/2}$, the error is integrable and contributes $\mathcal{O}_M(t^{2M})$. The interval is symmetric, P_ν has parity ν , and hence every odd coefficient integrates to zero. Extending the resulting Gaussian integrals to $\mathbb{R}$ changes them by $\mathcal{O}(\exp\{-ct^{-1/2}\})$.

For $|\theta| \leq \eta/r$, Taylor's theorem applied to $\operatorname{Re} \Phi_r(\theta)$, with η chosen small, yields the uniform quadratic bound

$$\operatorname{Re} \Phi_r(\theta) \leq -\frac{1}{4} A \theta^2. \quad (24)$$

Thus $\mathcal{C}_1$ contributes $\mathcal{O}(\exp\{-ct^{-1/2}\})$.

Finally,

$$\operatorname{Re} \Phi_r(\theta) = e^{r \cos \theta} \cos(r \sin \theta) - e^r \leq e^{r \cos \theta} - e^r. \quad (25)$$

For $\eta/r \leq |\theta| \leq \pi$, the inequality $1 - \cos \theta \geq 2\theta^2/\pi^2$ gives

$$\begin{aligned} \operatorname{Re} \Phi_r(\theta) &\leq -e^r (1 - \exp\{-r(1 - \cos \theta)\}) \\ &\leq -c \frac{e^r}{r} = -c \frac{n}{r^2}, \end{aligned}$$

which proves (22). Reinstating the Jacobian $d\theta = du/\sqrt{A}$ and the factor in (9) gives (20). Since $t^{2M} = q^M$ and all discarded contour pieces are smaller than every power of q , (21) follows. $\square$

Remark 3.2 (Genuine scale). *The expansion parameter is $q = r/n$, not $1/n$ with bounded coefficients. The coefficients $C_m(r)$ remain bounded as $r \rightarrow \infty$, whereas writing the same series directly in powers of $1/n$ produces coefficients of order r^m . This distinction is indispensable when stating a correct remainder.*

4 Combinatorial coefficient calculus

4.1 The Gaussian functional

Define the normalized Gaussian functional

$$\mathfrak{G}[p] := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2/2} p(u) du. \quad (26)$$

Thus

$$\mathfrak{G}[u^{2a}] = (2a-1)!!, \quad \mathfrak{G}[u^{2a+1}] = 0. \quad (27)$$

For $\ell \geq 1$, put

$$h_\ell(u; r) := \alpha_{\ell+2}(r)(iu)^{\ell+2}. \quad (28)$$

Define polynomials $P_\nu(u; r)$ by

$$\exp\left(\sum_{\ell \geq 1} h_\ell(u; r) t^\ell\right) = \sum_{\nu \geq 0} P_\nu(u; r) t^\nu. \quad (29)$$

Theorem 4.1 (Equivalent formulae for every coefficient). *For $m \geq 0$, the coefficient in Theorem 3.1 has each of the following finite descriptions.*

(i) Gaussian coefficient extraction:

$$C_m(r) = \mathfrak{G}[P_{2m}(u; r)] = \mathfrak{G}\left([t^{2m}] \exp\left\{\sum_{\ell \geq 1} h_\ell(u; r) t^\ell\right\}\right). \quad (30)$$

(ii) Complete exponential Bell polynomial:

$$C_m(r) = \frac{1}{(2m)!} \mathfrak{G}[Y_{2m}(1!h_1, 2!h_2, \dots, (2m)!h_{2m})], \quad (31)$$

where Y_j is the j th complete exponential Bell polynomial.

(iii) Triangular recurrence:

$$P_0 = 1, \quad P_\nu = \frac{1}{\nu} \sum_{\ell=1}^{\nu} \ell h_\ell P_{\nu-\ell}, \quad C_m = \mathfrak{G}[P_{2m}]. \quad (32)$$

(iv) Integer-partition sum: if $\lambda = (1^{m_1} 2^{m_2} \dots) \vdash 2m$ and $K = \ell(\lambda) = \sum_{j \geq 1} m_j$, then

$$C_m(r) = \sum_{\lambda \vdash 2m} \frac{(-1)^{m+K} (2m + 2K - 1)!!}{[r(1+r)]^{m+K}} \prod_{j \geq 1} \frac{T_{j+2}(r)^{m_j}}{m_j! ((j+2)!)^{m_j}}. \quad (33)$$

Equivalently, summing over nonnegative integers $k_1, \dots, k_{2m}$ constrained by $\sum_{\ell=1}^{2m} \ell k_\ell = 2m$,

$$C_m(r) = \sum_{\sum \ell k_\ell = 2m} \frac{(-1)^{m+K} (2m + 2K - 1)!!}{[r(1+r)]^{m+K}} \prod_{\ell=1}^{2m} \frac{T_{\ell+2}(r)^{k_\ell}}{k_\ell! ((\ell+2)!)^{k_\ell}}, \quad K = \sum k_\ell. \quad (34)$$

Proof. Equation (30) is the coefficient obtained by inserting (18) into the central integral in the proof of Theorem 3.1. The standard generating identity

$$\exp \left(\sum_{j \geq 1} x_j \frac{t^j}{j!} \right) = \sum_{\nu \geq 0} Y_\nu(x_1, \dots, x_\nu) \frac{t^\nu}{\nu!}$$

gives (31). Differentiating (29) with respect to t and comparing coefficients gives (32).

For the partition formula, choose k_ℓ copies of $h_\ell t^\ell$ from the exponential. The constraint is $\sum \ell k_\ell = 2m$. If $K = \sum k_\ell$, the resulting power of u is

$$\sum_{\ell \geq 1} (\ell + 2) k_\ell = 2m + 2K.$$

Consequently the power of i is $(-1)^{m+K}$ and the Gaussian moment is $(2m + 2K - 1)!!$. Substitution of (17) gives (34), which is the multiplicity notation for (33). $\square$

4.2 Rational structure and degree bounds

Theorem 4.2 (Denominator theorem). *For every $m \geq 0$ there is a polynomial $P_m(r) \in \mathbb{Q}[r]$ such that*

$$C_m(r) = \frac{P_m(r)}{r^m (1+r)^{3m}}, \quad \deg P_m \leq 4m. \quad (35)$$

In particular, C_m is regular for $r > 0$ and has a finite limit as $r \rightarrow \infty$.

Proof. Every $T_j(r)$ with $j \geq 1$ contains a factor r . In a summand of (33) with length K , these factors cancel r^K from $[r(1+r)]^{m+K}$, leaving a denominator $r^m (1+r)^{m+K}$. Since a partition of $2m$ has $K \leq 2m$, a common denominator is $r^m (1+r)^{3m}$.

After removing the factor r from each $T_{j+2}(r)$, the numerator degree in that summand is at most

$$\sum_{j \geq 1} (j+1) m_j = 2m + K.$$

Multiplication by $(1+r)^{2m-K}$ to reach the common denominator raises this by at most $2m - K$, giving degree at most $4m$. $\square$

Corollary 4.3 (Finite algorithm and arithmetic). *For a fixed m , computing C_m requires only the Touchard polynomials through index $2m + 2$ and the integer partitions of $2m$. All arithmetic can be carried out exactly over $\mathbb{Q}(r)$.*

Remark 4.4. *The number of summands in (33) is the partition number $p(2m)$, rather than the much larger number of terms obtained by naively expanding repeated derivatives. This is the main practical advantage of the partition organization.*

5 Explicit principal coefficients

The first coefficient follows from the two partitions 2 and 1 + 1:

$$C_1 = 3\alpha_4 - \frac{15}{2}\alpha_3^2. \quad (36)$$

After substitution of the Touchard polynomials, the first five coefficients are as follows.

$$C_0(r) = 1, \quad (37)$$

$$C_1(r) = -\frac{2r^4 + 9r^3 + 16r^2 + 6r + 2}{24r(1+r)^3}, \quad (38)$$

$$C_2(r) = \frac{4r^8 - 12r^7 - 143r^6 - 384r^5 - 588r^4 - 636r^3 + 100r^2 + 24r + 4}{1152r^2(1+r)^6}, \quad (39)$$

$$C_3(r) = \frac{1}{414720r^3(1+r)^9} \left(1112r^{12} + 12708r^{11} + 56082r^{10} + 118683r^9 \right. \\ \left. + 97176r^8 - 74394r^7 - 259046r^6 - 456444r^5 - 635424r^4 \right. \\ \left. + 115308r^3 + 39432r^2 + 10008r + 1112 \right), \quad (40)$$

$$C_4(r) = -\frac{1}{39813120r^4(1+r)^{12}} \left(9136r^{16} - 357408r^{15} - 5092600r^{14} - 29432904r^{13} \right. \\ \left. - 98499653r^{12} - 212369184r^{11} - 302729704r^{10} - 277448328r^9 \right. \\ \left. - 156972248r^8 - 52300272r^7 + 110053352r^6 + 302787360r^5 \right. \\ \left. + 198352r^4 + 2545056r^3 + 647456r^2 + 109632r + 9136 \right). \quad (41)$$

The apparently small coefficient 198352 of r^4 in (41) is not a typographical omission; it follows from the exact partition sum and was checked independently by recurrence and direct series expansion.

First practical truncations

With $q = r/n$, the one-, two-, and three-correction approximations are

$$B_n = L_n(r)\{1 + \mathcal{O}(q)\}, \\ B_n = L_n(r)\{1 + C_1(r)q + \mathcal{O}(q^2)\}, \\ B_n = L_n(r)\{1 + C_1(r)q + C_2(r)q^2 + \mathcal{O}(q^3)\}.$$

For numerical work, the first neglected term is a useful local diagnostic, but it is not by itself a rigorous global error bound once the series is taken near optimal truncation.

6 The Stirling interface

6.1 The limiting coefficients

Let $\hat{s}_m$ denote the reciprocal Stirling coefficients, defined by

$$\frac{1}{\Gamma(x+1)} \sim \frac{e^x x^{-x-1/2}}{\sqrt{2\pi}} \sum_{m \geq 0} \hat{s}_m x^{-m}. \quad (42)$$

Their ordinary generating series is the formal exponential

$$\sum_{m \geq 0} \hat{s}_m z^m = \exp \left(- \sum_{\nu \geq 1} \frac{B_{2\nu}}{2\nu(2\nu-1)} z^{2\nu-1} \right), \quad (43)$$

where $B_{2\nu}$ are Bernoulli numbers.

Theorem 6.1 (Reciprocal-Stirling limit). *For every fixed $m \geq 0$,*

$$\lim_{r \rightarrow \infty} C_m(r) = \hat{s}_m. \quad (44)$$

In particular,

$$1, \quad -\frac{1}{12}, \quad \frac{1}{288}, \quad \frac{139}{51840}, \quad -\frac{571}{2488320}, \dots \quad (45)$$

are the limits of $C_0, C_1, C_2, C_3, C_4, \dots$

Proof. For fixed j ,

$$\alpha_j(r) = \frac{T_j(r)}{j![r(1+r)]^{j/2}} \rightarrow \frac{1}{j!}.$$

Hence (30) tends coefficientwise to

$$\mathfrak{G} \left([t^{2m}] \exp \left\{ \sum_{j \geq 3} \frac{(iu)^j}{j!} t^{j-2} \right\} \right). \quad (46)$$

To identify these numbers, use the exact diagonal integral

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \exp(xe^{i\theta} - x - ix\theta) d\theta = \frac{e^{-x} x^x}{\Gamma(x+1)} \quad (47)$$

for positive integer x , and apply the same Gaussian scaling $\theta = u/\sqrt{x}$. Stirling's expansion of the right-hand side identifies (46) with $\hat{s}_m$. Since both sides are formal coefficient identities, the conclusion holds for every m . $\square$

Remark 6.2. *Hwang proved a broader formal identity showing that two apparently different saddle coefficient systems both reduce to the same Stirling coefficients. Here only the specialization needed for the Bell circle saddle is used.*

6.2 Factorial-free all-order expansion

Let the ordinary Stirling coefficients s_j be defined by

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e} \right)^n \sum_{j \geq 0} s_j n^{-j}, \quad (48)$$

with

$$\sum_{j \geq 0} s_j z^j = \exp \left(\sum_{\nu \geq 1} \frac{B_{2\nu}}{2\nu(2\nu-1)} z^{2\nu-1} \right). \quad (49)$$

Thus $s_0 = 1$, $s_1 = 1/12$, $s_2 = 1/288$, $s_3 = -139/51840$, and so on.

Theorem 6.3 (Factorial-free form). *Put*

$$D_m(r) := r^m C_m(r), \quad \mathcal{E}_\ell(r) := \sum_{m=0}^{\ell} s_{\ell-m} D_m(r). \quad (50)$$

Then

$$B_n \sim \frac{\exp\{nr - n + n/r - 1\}}{\sqrt{1+r}} \sum_{\ell \geq 0} \frac{\mathcal{E}_\ell(r)}{n^\ell}, \quad r = W_0(n). \quad (51)$$

Although written in powers of $1/n$, this is still an expansion in the asymptotic scale $(r/n)^\ell$, because $\mathcal{E}_\ell(r) = \mathcal{O}(r^\ell)$.

Proof. Insert (48) into (20) and use $\log(n/r) = r$ and $e^r = n/r$. Cauchy multiplication gives (50). The growth statement follows from Theorem 6.1. $\square$

The first coefficients in this normalization are

$$\mathcal{E}_0(r) = 1, \quad (52)$$

$$\mathcal{E}_1(r) = -\frac{r^2(2r^2 + 7r + 10)}{24(1+r)^3}, \quad (53)$$

$$\mathcal{E}_2(r) = \frac{r^3(4r^5 - 20r^4 - 199r^3 - 556r^2 - 860r - 864)}{1152(1+r)^6}, \quad (54)$$

$$\begin{aligned} \mathcal{E}_3(r) = \frac{r^4}{414720(1+r)^9} & \left(1112r^8 + 12828r^7 + 55962r^6 + 111301r^5 \right. \\ & \left. + 55818r^4 - 196476r^3 - 492584r^2 - 749376r - 867456 \right). \end{aligned} \quad (55)$$

The negative sign in (53) is important: it follows from a cancellation between the $+1/(12n)$ Stirling correction and the larger saddle correction $rC_1(r)/n$.

7 The logarithmic transseries

Multiplicative asymptotic series are often best manipulated after taking a formal logarithm. Define

$$F(q; r) := 1 + \sum_{m \geq 1} C_m(r) q^m, \quad \log F(q; r) = \sum_{m \geq 1} \Lambda_m(r) q^m. \quad (56)$$

Proposition 7.1 (Cumulant recurrence). *The coefficients Λ_m are determined by*

$$\Lambda_m(r) = C_m(r) - \frac{1}{m} \sum_{j=1}^{m-1} j \Lambda_j(r) C_{m-j}(r), \quad m \geq 1. \quad (57)$$

Thus

$$\begin{aligned} \Lambda_1 &= C_1, \\ \Lambda_2 &= C_2 - \frac{1}{2} C_1^2, \\ \Lambda_3 &= C_3 - C_1 C_2 + \frac{1}{3} C_1^3. \end{aligned}$$

Proof. Differentiate $F = \exp(\log F)$, use $F' = F(\log F)'$, and compare the coefficient of q^{m-1} . $\square$

The first two nonzero cumulants simplify to

$$\Lambda_1(r) = -\frac{2r^4 + 9r^3 + 16r^2 + 6r + 2}{24r(1+r)^3}, \quad (58)$$

$$\Lambda_2(r) = -\frac{r(2r^4 + 12r^3 + 29r^2 + 40r + 36)}{48(1+r)^6}. \quad (59)$$

Theorem 7.2 (All-order logarithm). *For fixed truncation order, the logarithm of the Bell number has the expansion*

$$\begin{aligned} \log B_n \sim & n \left(r - 1 + \frac{1}{r} \right) - 1 - \frac{1}{2} \log(1+r) \\ & + \sum_{\nu \geq 1} \frac{B_{2\nu}}{2\nu(2\nu-1)n^{2\nu-1}} + \sum_{m \geq 1} \Lambda_m(r) \left(\frac{r}{n} \right)^m, \quad r = W_0(n). \end{aligned} \quad (60)$$

Equivalently,

$$\log B_n \sim nA(r) + B(r) + \sum_{j \geq 1} \frac{H_j(r)}{n^j}, \quad (61)$$

where

$$A(r) := r - 1 + r^{-1}, \quad (62)$$

$$B(r) := -1 - \frac{1}{2} \log(1+r), \quad (63)$$

$$H_j(r) := r^j \Lambda_j(r) + \mathbf{1}_{\{j \text{ odd}\}} \frac{B_{j+1}}{j(j+1)}. \quad (64)$$

Every H_j is an explicitly computable rational function of r .

Proof. Take the logarithm of (20), use the Stirling expansion for $\log n!$, and simplify with $\log(n/r) = r$ and $e^r = n/r$. The formal logarithm of the correction factor is (56). Formula (64) only regroups terms carrying the same power of $1/n$. $\square$

Remark 7.3 (Residual certificates). *Suppose $F_M = 1 + \sum_{m=1}^{M-1} C_m q^m$. Formal division verifies*

$$\exp \left(\sum_{m=1}^{M-1} \Lambda_m q^m \right) - F_M = \mathcal{O}(q^M).$$

This identity is a simple symbolic certificate that the multiplicative and logarithmic truncations agree through the requested order.

8 Lambert- W polynomial-logarithmic re-expansion

8.1 Every branch

Let z tend to infinity away from the branch cuts and put

$$\xi_k := \text{Log } z + 2\pi i k. \quad (65)$$

Let $\left[\begin{smallmatrix} a \\ b \end{smallmatrix} \right]$ denote an unsigned Stirling number of the first kind. Define the Lambert polynomials

$$\mathcal{L}_a(y) := \sum_{b=1}^a (-1)^{a-b} \left[\begin{smallmatrix} a \\ a-b+1 \end{smallmatrix} \right] \frac{y^b}{b!}, \quad a \geq 1. \quad (66)$$

The standard branch expansion can then be written compactly as

$$W_k(z) \sim \xi_k - \log \xi_k + \sum_{a \geq 1} \frac{\mathcal{L}_a(\log \xi_k)}{\xi_k^a}. \quad (67)$$

This formula supplies every coefficient using only Stirling numbers of the first kind. The first polynomials are

$$\begin{aligned} \mathcal{L}_1(y) &= y, \\ \mathcal{L}_2(y) &= \frac{y^2}{2} - y, \\ \mathcal{L}_3(y) &= \frac{y^3}{3} - \frac{3}{2}y^2 + y, \\ \mathcal{L}_4(y) &= \frac{y^4}{4} - \frac{11}{6}y^3 + 3y^2 - y, \\ \mathcal{L}_5(y) &= \frac{y^5}{5} - \frac{25}{12}y^4 + \frac{35}{6}y^3 - 5y^2 + y. \end{aligned}$$

For the principal real branch, write

$$L := \log n, \quad \ell := \log L. \quad (68)$$

Then

$$\begin{aligned} r = W_0(n) \sim L - \ell + \frac{\ell}{L} + \frac{\ell(\ell-2)}{2L^2} + \frac{\ell(6-9\ell+2\ell^2)}{6L^3} \\ + \frac{\ell(-12+36\ell-22\ell^2+3\ell^3)}{12L^4} + \dots \end{aligned} \quad (69)$$

8.2 Canonical nested-log form of $\log B_n$

Substitution of (69) into (60) produces the polynomial-logarithmic transseries without any further inversion. The first inverse-log layers, before the algebraic $1/n$ layer, are

$$\begin{aligned} \log B_n \sim n \left[L - \ell - 1 + \frac{\ell+1}{L} + \frac{\ell^2}{2L^2} + \frac{\ell^3/3 - \ell^2/2}{L^3} \right. \\ \left. + \frac{\ell^4/4 - 5\ell^3/6 + \ell^2/2}{L^4} + \dots \right] \\ - \frac{1}{2} \log L - 1 + \frac{\ell-1}{2L} + \frac{\ell^2-4\ell+1}{4L^2} \\ + \frac{2\ell^3-15\ell^2+18\ell-2}{12L^3} + \dots + \sum_{j \geq 1} \frac{H_j(W_0(n))}{n^j}. \end{aligned} \quad (70)$$

The final sum in (70) is itself expanded by (69). It starts at the much smaller scale $\log n/n$.

Algorithm 8.1 (Mechanical nested-log expansion). To compute (70) through any prescribed monomial order:

1. generate $\mathcal{L}_a$ from (66);
2. truncate (67) at the required inverse-log order;
3. generate C_m from (33), then Λ_m from (57);

4. substitute the truncated W series into A, B, H_j in (61);

5. expand in the ordered monomial family $n^{-j} L^{-a} \ell^b$, retaining the chosen finite lower set.

Because every step is a finite polynomial or rational operation at a fixed order, the procedure terminates and gives a residual certificate by direct substitution.

9 The full Lambert-branch saddle transseries

The principal circle expansion cannot see exponentially small saddles one at a time. For that purpose we return to the Hankel integral (13).

9.1 Local data at an arbitrary saddle

At $w = w_k = W_k(N)$,

$$\psi_N(w) = \frac{1}{w} - \text{Log } w, \quad \psi_N''(w) = \frac{1+w}{w^2}, \quad (71)$$

and for $j \geq 2$,

$$\psi_N^{(j)}(w) = \frac{1}{w} + (-1)^j \frac{(j-1)!}{w^j}. \quad (72)$$

Use the local steepest coordinate

$$t = w + \frac{iw}{\sqrt{N(1+w)}} u. \quad (73)$$

Define

$$\beta_j(w) := \frac{w^j \psi_N^{(j)}(w)}{j!(1+w)^{j/2}} = \frac{w^{j-1} + (-1)^j (j-1)!}{j!(1+w)^{j/2}}, \quad j \geq 3. \quad (74)$$

Then the local exponent is

$$N\{\psi_N(t) - \psi_N(w)\} = -\frac{u^2}{2} + \sum_{j \geq 3} \beta_j(w) (iu)^j N^{-(j-2)/2}. \quad (75)$$

9.2 All coefficients in one thimble sector

For $\ell \geq 1$, set

$$\tilde{h}_\ell(u; w) := \beta_{\ell+2}(w) (iu)^{\ell+2}. \quad (76)$$

Define $\tilde{P}_\nu$ by the same triangular recurrence as before:

$$\tilde{P}_0 = 1, \quad \tilde{P}_\nu = \frac{1}{\nu} \sum_{\ell=1}^{\nu} \ell \tilde{h}_\ell \tilde{P}_{\nu-\ell}. \quad (77)$$

Theorem 9.1 (Arbitrary Lambert-sector expansion). *For a fixed saddle branch k whose thimble contributes, let $w_k = W_k(N)$. Its oriented contribution has the formal Poincaré expansion*

$$\mathcal{J}_k(N) \sim \frac{\Gamma(N) \exp(-1 + N/w_k)}{w_k^{N-1} \sqrt{2\pi N(1+w_k)}} \sum_{m \geq 0} \frac{\tilde{C}_m(w_k)}{N^m}, \quad (78)$$

where the square root is fixed by the thimble orientation and

$$\tilde{C}_m(w) = \mathfrak{G}[\tilde{P}_{2m}(u; w)]. \quad (79)$$

Equivalently,

$$\tilde{C}_m(w) = \sum_{\lambda \vdash 2m} (-1)^{m+K} (2m + 2K - 1)!! \prod_{j \geq 1} \frac{\beta_{j+2}(w)^{m_j}}{m_j!}, \quad K = \ell(\lambda). \quad (80)$$

Moreover,

$$\tilde{C}_m(w) = \frac{\tilde{P}_m(w)}{(1+w)^{3m}}, \quad \tilde{P}_m \in \mathbb{Q}[w], \quad \deg \tilde{P}_m \leq 4m. \quad (81)$$

Proof. Insert (73) into the thimble integral and use (75). The Gaussian coefficient extraction and partition expansion are identical to the proof of Theorem 4.1, with $q^{1/2}$ replaced by $N^{-1/2}$ and α_j replaced by β_j . Each factor β_{j+2} has denominator $(1+w)^{(j+2)/2}$. A partition of $2m$ of length K therefore contributes $(1+w)^{-(m+K)}$; since $K \leq 2m$, the common denominator is $(1+w)^{3m}$. The numerator-degree estimate is the same as in Theorem 4.2. $\square$

The first coefficients are

$$\tilde{C}_0(w) = 1, \quad (82)$$

$$\tilde{C}_1(w) = -\frac{2w^4 - 3w^3 - 20w^2 - 18w + 2}{24(1+w)^3}, \quad (83)$$

$$\tilde{C}_2(w) = \frac{4w^8 - 156w^7 - 695w^6 - 696w^5 + 1092w^4 + 2916w^3 + 1972w^2 - 72w + 4}{1152(1+w)^6}. \quad (84)$$

In Paris's notation these are $\tilde{C}_m = c_{2m}(1/2)_m$; thus (83)–(84) agree with the published low-order saddle coefficients, while (80) supplies all higher orders.

9.3 Thimble selection and Stokes multipliers

For positive real argument, the saddles occur in conjugate pairs. Deforming the original Hankel loop into steepest-descent paths gives the exact contour decomposition

$$B_n = \mathcal{J}_0(N) + 2 \operatorname{Re} \sum_{k=1}^{K(N)} \mathcal{J}_k(N), \quad N = n + 1, \quad (85)$$

away from a Stokes threshold. The integer $K(N)$ is determined by thimble connectivity. It increases when the positive-axis parameter N crosses a Stokes boundary. The first boundaries affecting the positive axis are

range of N	active positive indices	total saddle count
$0 < N < 6.87877 \dots$	$k = 1$	3
$6.87877 \dots < N < 13.56411 \dots$	$k = 1, 2$	5
$13.56411 \dots < N < 20.13877 \dots$	$k = 1, 2, 3$	7
$20.13877 \dots < N < 26.43594 \dots$	$k = 1, 2, 3, 4$	9

The omitted boundaries continue by the same adjacent-thimble Stokes test. A computational definition is: a new pair is switched on when adjacent saddle actions have equal imaginary part,

$$\operatorname{Im}\{\psi_N(w_j) - \psi_N(w_{j+1})\} = 0, \quad (86)$$

and the corresponding steepest path has the required connectivity to the original loop. The connectivity clause is essential; equality of phases alone does not determine an intersection number.

It is convenient to encode this by Stokes multipliers $\sigma_k(N) \in \{0, 1\}$ away from a transition:

$$B_n \sim \mathcal{J}_0^{\text{formal}}(N) + 2 \operatorname{Re} \sum_{k \geq 1} \sigma_k(N) \mathcal{J}_k^{\text{formal}}(N), \quad (87)$$

where $\mathcal{J}_k^{\text{formal}}$ denotes (78). At every fixed N only finitely many σ_k are nonzero. A median convention gives $\sigma_k = 1/2$ for a sector exactly on its Stokes line; lateral contours give the two one-sided values.

Remark 9.2 (Meaning of the global formula). *Equation (85) is an exact contour statement. Equation (87) attaches a Poincaré series to each exact thimble. The statement is sectorwise, not a claim that all active series may be truncated at one common order with an error estimate uniform in k as $K(N) \rightarrow \infty$.*

9.4 Relation between the two principal normalizations

The $k = 0$ sector in (78) uses $w_0 = W_0(n + 1)$ and a $1/(n + 1)$ expansion. The circle result uses $r = W_0(n)$ and $q = r/n$. Both are expansions of the same exact quantity, but their coefficient functions are not identical term by term. They are related by the formal shift

$$w_0 = r + \frac{r}{n(1+r)} + \mathcal{O}\left(\frac{r^2}{n^2}\right) \quad (88)$$

and subsequent re-expansion. Keeping the two forms separate makes each local calculus simpler and prevents accidental mixing of n and $n + 1$.

10 Singulants and beyond-all-orders ordering

10.1 Exact singulant

Let $R := w_0 = W_0(N)$. Compare the leading term of the k th sector with the principal one. Define

$$\mathcal{A}_k(N) := N \left\{ \operatorname{Log} \frac{w_k}{R} + \frac{1}{R} - \frac{1}{w_k} \right\}. \quad (89)$$

Then

$$\frac{\mathcal{J}_k^{(0)}(N)}{\mathcal{J}_0^{(0)}(N)} = \exp\{-\mathcal{A}_k(N)\} \frac{w_k}{R} \sqrt{\frac{1+R}{1+w_k}}. \quad (90)$$

The logarithms in (89) are chosen by continuation from the corresponding W branches, consistently with the thimble orientation.

10.2 Fixed-branch expansion

Set

$$a_k := 2\pi i k, \quad \delta_k := w_k - R. \quad (91)$$

Subtracting $R + \log R = \log N$ from $w_k + \operatorname{Log} w_k = \log N + 2\pi i k$ gives the exact implicit equation

$$\delta_k + \log \left(1 + \frac{\delta_k}{R} \right) = a_k. \quad (92)$$

Its first terms are

$$\begin{aligned} \delta_k = a_k - \frac{a_k}{R} + \frac{a_k(a_k + 2)}{2R^2} - \frac{a_k(2a_k^2 + 9a_k + 6)}{6R^3} \\ + \frac{a_k(3a_k^3 + 22a_k^2 + 36a_k + 12)}{12R^4} + \mathcal{O}_k(R^{-5}). \end{aligned} \quad (93)$$

All coefficients are generated by the finite recurrence below. If $\delta(a, x) = \sum_{m \geq 0} d_m(a) x^m$ with $x = R^{-1}$ and $d_0(a) = a$, then

$$d_m(a) = - \sum_{j=1}^m \frac{(-1)^{j+1}}{j} [x^{m-j}] \delta(a, x)^j, \quad m \geq 1. \quad (94)$$

Substitution into (89) yields

$$\begin{aligned} \frac{\mathcal{A}_k(N)}{N} &= \frac{a_k}{R} - \frac{a_k^2}{2R^2} + \frac{a_k^3/3 + a_k^2/2}{R^3} \\ &\quad - \frac{a_k^4/4 + 5a_k^3/6 + a_k^2/2}{R^4} + \mathcal{O}_k(R^{-5}). \end{aligned} \quad (95)$$

In particular,

$$\operatorname{Re} \mathcal{A}_k(N) = \frac{2\pi^2 k^2 N}{R^2} \left(1 - \frac{1}{R} + \mathcal{O}_k(R^{-2}) \right). \quad (96)$$

Corollary 10.1 (Beyond every algebraic order). *For each fixed $k \neq 0$ and every fixed $M > 0$,*

$$\frac{\mathcal{J}_k(N)}{\mathcal{J}_0(N)} = o\left(\left(\frac{R}{N}\right)^M\right) \quad (N \rightarrow \infty), \quad (97)$$

sectorwise after fixed-order truncation. More explicitly, the leading magnitude is

$$\left| \frac{\mathcal{J}_k^{(0)}}{\mathcal{J}_0^{(0)}} \right| = \exp\left\{ -\frac{2\pi^2 k^2 N}{R^2} (1 + \mathcal{O}_k(R^{-1})) \right\} \mathcal{O}_k(1). \quad (98)$$

Proof. Equation (96) shows that the logarithm of the sector ratio is $-\Theta_k(N/\log^2 N)$, whereas the logarithm of any fixed power of R/N is only $-\Theta_M(\log N)$. Therefore the exponential sector is smaller. $\square$

Remark 10.2 (Oscillations). *The imaginary part of $\mathcal{A}_k$ begins with $2\pi k N/R - 8\pi^3 k^3 N/(3R^3) + \dots$. The conjugate pair $k, -k$ consequently contributes an exponentially damped cosine with a rapidly varying phase. The exact phase also includes the argument of the prefactor in (90) and the phase of the truncated coefficient series.*

11 Executable coefficient generation

The following compact SymPy program implements the partition formula exactly. It generates C_m , verifies the denominator theorem, and also generates the arbitrary-thimble coefficients $\tilde{C}_m$.

Listing 1: Exact principal and thimble coefficient generation.

```
import sympy as sp

r, w = sp.symbols("r w")

def touchard(j, x):
    return sp.expand(sum(
        sp.functions.combinatorial.numbers.stirling(j, a, kind=2) * x**a
        for a in range(j + 1)
    ))
```

```

def multiplicity_partitions(total):
    """Yield dictionaries {part: multiplicity} for partitions of total."""
    def rec(rem, part, data):
        if rem == 0:
            yield dict(data)
            return
        if part == 0:
            return
        for count in range(rem // part, -1, -1):
            if count:
                data[part] = count
            else:
                data.pop(part, None)
            yield from rec(rem - count * part, part - 1, data)
        data.pop(part, None)
    yield from rec(total, total, {})

def principal_C(m):
    if m == 0:
        return sp.Integer(1)
    ans = 0
    for mult in multiplicity_partitions(2 * m):
        K = sum(mult.values())
        term = (-1)**(m + K) * sp.factorial2(2 * m + 2 * K - 1)
        term /= (r * (1 + r))**(m + K)
        for ell, count in mult.items():
            term *= touchard(ell + 2, r)**count
            term /= sp.factorial(count) * sp.factorial(ell + 2)**count
        ans += term
    return sp.factor(sp.cancel(ans))

def beta(j):
    return (w**(j - 1) + (-1)**j * sp.factorial(j - 1)) / (
        sp.factorial(j) * (1 + w)**sp.Rational(j, 2)
    )

def thimble_C(m):
    if m == 0:
        return sp.Integer(1)
    ans = 0
    for mult in multiplicity_partitions(2 * m):
        K = sum(mult.values())
        term = (-1)**(m + K) * sp.factorial2(2 * m + 2 * K - 1)
        for ell, count in mult.items():
            term *= beta(ell + 2)**count / sp.factorial(count)
        ans += term
    return sp.factor(sp.cancel(ans))

for m in range(5):
    Cm = principal_C(m)
    print(m, Cm)
    print("denominator check:",
          sp.cancel(Cm * r**m * (1 + r)**(3*m)))

for m in range(3):
    print("thimble", m, thimble_C(m))

```

11.1 Stable numerical evaluation

For large n , do not form $n!$ or B_n directly in floating-point arithmetic. Compute

$$\log L_n(r) = \log \Gamma(n+1) + e^r - 1 - n \log r - \frac{1}{2} \log\{2\pi n(1+r)\} \quad (99)$$

and evaluate the correction polynomial in q by Horner's rule. For a signed relative approximation, return

$$\widehat{B}_n^{(M)} = \exp\{\log L_n(r)\} \sum_{m=0}^{M-1} C_m(r) q^m. \quad (100)$$

When only $\log B_n$ is required, use (60) directly; this is both more stable and more naturally compatible with transseries composition.

12 Numerical validation

Let $\widehat{B}_n^{(M)}$ denote (100), with M terms $C_0, \dots, C_{M-1}$, and define the signed relative error

$$\varepsilon_{n,M} := \frac{\widehat{B}_n^{(M)}}{B_n} - 1. \quad (101)$$

The exact Bell numbers in the following table were generated by the integer recurrence

$$B_0 = 1, \quad B_n = \sum_{j=0}^{n-1} \binom{n-1}{j} B_j. \quad (102)$$

n	$M = 1$	$M = 2$	$M = 3$	$M = 4$	$M = 5$
10	$2.6771612450 \cdot 10^{-2}$	$3.8253602751 \cdot 10^{-4}$	$-1.2965341197 \cdot 10^{-5}$	$-6.2268397638 \cdot 10^{-6}$	$-1.1599280809 \cdot 10^{-6}$
20	$1.5326827723 \cdot 10^{-2}$	$1.1293363992 \cdot 10^{-4}$	$-3.8237416551 \cdot 10^{-6}$	$-7.5548760506 \cdot 10^{-7}$	$-7.1302490456 \cdot 10^{-8}$
50	$7.2381177878 \cdot 10^{-3}$	$2.1197164537 \cdot 10^{-5}$	$-5.8160606659 \cdot 10^{-7}$	$-4.1043863046 \cdot 10^{-8}$	$-1.3992028317 \cdot 10^{-9}$
100	$4.0615707492 \cdot 10^{-3}$	$5.7437480134 \cdot 10^{-6}$	$-1.2399465806 \cdot 10^{-7}$	$-4.2053807468 \cdot 10^{-9}$	$-6.2657057618 \cdot 10^{-11}$
200	$2.2599385476 \cdot 10^{-3}$	$1.5057676926 \cdot 10^{-6}$	$-2.4589871384 \cdot 10^{-8}$	$-4.0769478428 \cdot 10^{-10}$	$-2.4569359891 \cdot 10^{-12}$
500	$1.0289535458 \cdot 10^{-3}$	$2.4364162415 \cdot 10^{-7}$	$-2.6649303424 \cdot 10^{-9}$	$-1.7364703016 \cdot 10^{-11}$	$-2.4871987030 \cdot 10^{-14}$

Several features are visible. The leading approximation is already within a few percent at $n = 10$. The C_1 correction improves it by roughly two orders of magnitude. The signs need not alternate monotonically at small n , but fixed-order accuracy rapidly approaches the predicted q^M scale. The five-term error at $n = 500$ is at the level of 2.5×10^{-14} in the working precision used for the calculation.

13 Proof-audit and formalization map

This section records the logical dependencies in a form suitable for later formalization.

13.1 Analytic kernel

A proof assistant development would require the following analytic statements.

1. Cauchy's coefficient formula specialized to the circle $z = re^{i\theta}$, giving (9).
2. Existence and uniqueness of $r > 0$ satisfying $re^r = n$.
3. The derivative identity (8) and the exact phase expansion (15).
4. Uniform normalized derivative bounds such as (19).

5. The local quadratic decay (24) and the global bound (25).
 6. A finite Taylor theorem under a Gaussian integral, with a remainder uniform in r once $r \geq 1$.
- These yield Theorem 3.1 without invoking any unformalized general “saddle-point method” black box.

13.2 Algebraic kernel

The coefficient calculus reduces to finite combinatorics:

1. exponential Bell polynomial generating identities;
2. the recurrence (32);
3. Gaussian moments (27);
4. the bijection between constrained multiplicity vectors and partitions;
5. polynomial degree and denominator bookkeeping.

This part is particularly well suited to exact symbolic verification.

13.3 Transseries kernel

The global extension additionally needs:

1. the Hankel representation (13);
2. the branch equation and conjugacy properties of W_k ;
3. oriented thimble deformation and intersection numbers;
4. local saddle expansion on each thimble;
5. a formal algebra for the branch expansion (67) and for composition in the monomial family

$$n^{-j} L^{-a} \ell^b e^{-A_k}.$$

The present article separates these analytic, algebraic, and transseries layers so that no formal identity is mistaken for a uniform analytic theorem.

14 Conclusions

The Bell numbers admit two complementary complete descriptions.

First, on the principal Cauchy circle, they possess a rigorous fixed-order expansion in $q = W_0(n)/n$. Every coefficient is a finite sum over partitions of an even integer; its denominator and degree are controlled uniformly in the order. This resolves the usual ambiguity hidden behind phrases such as “higher coefficients may be computed.” The same formulas lead immediately to factorial-free, logarithmic, and nested-log forms.

Second, the Hankel contour resolves the beyond-all-orders structure. Every Lambert branch $W_k(n+1)$ gives a thimble sector with its own all-order coefficient series. The Bell contour fixes which sectors are active, and the singulants order them exponentially. For fixed $k \neq 0$, the suppression $\exp\{-2\pi^2 k^2 n / \log^2 n\}$ explains why no finite algebraic expansion can recover these oscillatory corrections.

Together, the principal partition calculus, the Lambert-polynomial branch expansions, and the thimble sectors form a complete saddle-transseries framework. They also supply direct algorithms for extending the displayed coefficients to any desired order without introducing unnamed recurrences or undocumented shifts.

A A sharper view of the contour remainder

This appendix isolates the two inequalities used in Theorem 3.1.

Lemma A.1 (Uniform local domination). *There are absolute constants $\eta, c_0 > 0$ and r_0 such that, whenever $r \geq r_0$, $n = re^r$, and $|\theta| \leq \eta/r$,*

$$\operatorname{Re} \Phi_r(\theta) \leq -c_0 n(1+r)\theta^2. \quad (103)$$

Proof. The real part is an even analytic function of θ , with value and first derivative zero and second derivative $-A = -n(1+r)$. For every fixed $j \geq 2$, the $2j$ th derivative at the origin is bounded in magnitude by $e^r T_{2j}(r) \leq e^r B_{2j} r^{2j}$ for $r \geq 1$. If $|\theta| \leq \eta/r$, the sum of all normalized terms of order at least four is bounded by $C\eta^2 A\theta^2$ for a sufficiently small universal η ; this can also be seen directly from analyticity of $e^{re^{i\theta}}$ in the disk $|r\theta| < 2\eta$. Taking $C\eta^2 \leq 1/4$ gives the result, for example with $c_0 = 1/4$. $\square$

Lemma A.2 (Outer-arc domination). *For the same fixed $\eta > 0$, there is $c_1 > 0$ such that*

$$\sup_{\eta/r \leq |\theta| \leq \pi} \operatorname{Re} \Phi_r(\theta) \leq -c_1 \frac{n}{r^2} \quad (104)$$

for all sufficiently large r .

Proof. From (25) and $1 - \cos \theta \geq 2\theta^2/\pi^2$,

$$\operatorname{Re} \Phi_r(\theta) \leq -e^r \left(1 - \exp \left\{ -\frac{2r\theta^2}{\pi^2} \right\} \right).$$

On the stated arc the exponent is at least $2\eta^2/(\pi^2 r)$. Since $1 - e^{-x} \geq x/2$ for small positive x , the right-hand side is at most $-ce^r/r = -cn/r^2$. $\square$

Proposition A.3 (Two-scale remainder). *For every fixed M ,*

$$R_M(n) = \mathcal{O}_M(q^M) + \mathcal{O}\left(e^{-cq^{-1/4}}\right) + \mathcal{O}\left(e^{-cn/r^2}\right). \quad (105)$$

The two exponential terms are $o(q^J)$ for every fixed J .

Proof. The first term is the integrated Taylor remainder on $|u| \leq q^{-1/8}$ (equivalent to $|u| \leq t^{-1/4}$ above). The second is the Gaussian tail on the remainder of the local arc, and the third follows from Theorem A.2. Since $q^{-1/4} \rightarrow \infty$ faster than $\log(1/q)$ and $n/r^2 \gg \log(1/q)$, both exponentials are beyond all powers of q . $\square$

B Coefficient identities in expanded form

For reference, the coefficient extractor may be written without auxiliary polynomials as

$$C_m(r) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2/2} [t^{2m}] \exp \left(\sum_{j \geq 3} \frac{T_j(r)(iu)^j}{j! [r(1+r)]^{j/2}} t^{j-2} \right) du. \quad (106)$$

Likewise,

$$\tilde{C}_m(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2/2} [t^{2m}] \exp \left(\sum_{j \geq 3} \frac{w^{j-1} + (-1)^j(j-1)!}{j!(1+w)^{j/2}} (iu)^j t^{j-2} \right) du. \quad (107)$$

At a fixed order, each coefficient extraction contains only finitely many j ; the infinite sums in (106) and (107) are formal shorthand.

The reciprocal Stirling coefficients satisfy the useful recurrence

$$\widehat{s}_0 = 1, \quad \widehat{s}_m = \frac{1}{m} \sum_{j=1}^m j a_j \widehat{s}_{m-j}, \quad (108)$$

where

$$a_j = \begin{cases} -\frac{B_{j+1}}{j(j+1)}, & j \text{ odd,} \\ 0, & j \text{ even.} \end{cases} \quad (109)$$

The ordinary Stirling coefficients satisfy the same recurrence with the opposite sign in (109).

C Notation concordance

symbol	meaning
B_n	n th Bell number
$T_n(x)$	Touchard polynomial, so $B_n = T_n(1)$
r	principal circle saddle $W_0(n)$
q	principal small parameter $r/n = e^{-r}$
A	circle curvature $n(1+r)$
$C_m(r)$	principal multiplicative saddle coefficient
$P_m(r)$	numerator in $C_m = P_m / \{r^m(1+r)^{3m}\}$
$\Lambda_m(r)$	logarithmic/cumulant coefficient
$s_m, \widehat{s}_m$	ordinary and reciprocal Stirling coefficients
N	shifted parameter $n+1$ in the Hankel representation
w_k	Lambert saddle $W_k(N)$
$\widetilde{C}_m(w)$	arbitrary-thimble coefficient
$\mathcal{J}_k$	oriented contribution of the k th thimble
$\mathcal{A}_k$	singulant comparing sector k with sector 0
L, ℓ	$\log n$ and $\log \log n$

References

- [1] E. T. Bell, Exponential polynomials, *Annals of Mathematics* **35** (1934), 258–277.
- [2] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Advances in Computational Mathematics* **5** (1996), 329–359.
- [3] N. G. de Bruijn, *Asymptotic Methods in Analysis*, 3rd ed., Dover, New York, 1981.
- [4] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, Cambridge, 2009.
- [5] J. Grunwald and G. Serafin, Explicit bounds for Bell numbers and their ratios, *Journal of Mathematical Analysis and Applications* **549** (2025), Article 129527; arXiv:2408.14182.
- [6] W. K. Hayman, A generalisation of Stirling’s formula, *Journal für die reine und angewandte Mathematik* **196** (1956), 67–95.

- [7] H.-K. Hwang, A curious identity arising from Stirling's formula and saddle-point method on two different contours, *Electronic Journal of Combinatorics* **30** (2023), Paper P4.3.
- [8] L. Moser and M. Wyman, An asymptotic formula for the Bell numbers, *Transactions of the Royal Society of Canada, Section III* **49** (1955), 49–54.
- [9] NIST Digital Library of Mathematical Functions, §4.13, Lambert W -function, <https://dlmf.nist.gov/4.13>.
- [10] R. B. Paris, The asymptotics of the Touchard polynomials, *Mathematica Aeterna* **6** (2016), 765–779; arXiv:1606.07883.
- [11] J. Wojdylo, Computing the coefficients in Laplace's method, *SIAM Review* **48** (2006), 76–96.
- [12] V. Reshetnikov, *Combinatorial Coefficient Calculus*, ProveIt research draft, ProveIt repository path: [combinatorial-coefficient-calculus](#), accessed September 2026.
- [13] V. Reshetnikov, *Series and Transseries*, ProveIt research drafts, ProveIt repository path: [series-and-transseries](#), accessed September 2026.